INCH-POUND
MIL-DTL-46735C(AT)
31 August 1998
SUPERSEDING
MIL-PRF-46735B(AT)
26 March 1997

DETAIL SPECIFICATION

WINCH DRUM: VEHICLE MOUNTING

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for power operated spur planetary and planetary geared, heavy duty, drum type winches for use on military combat and tactical transport vehicles. The winches are driven from the vehicle hydraulic system.

1.2 Classification. Winches will be of the following models, classes and capacities including the drawing numbers as specified in table I (see 6.2).

TABLE I. Winch model, class and capacity.

Model number	Drawing number	Class		Capacity (tons)				
		Spur gear	Planetary gear					
		1	2	12.5	15	17.5	30	45
U-35B	8738935	X				X		
U-60B	8738936	X					X	
U-90B	8738937	X						X
P-30	10908544		X		X			
P-60	10904960		X				X	
N/A	11641136	X		X				

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2590

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

8738935	- Winch, U-35B.
8738936	- Winch, U-60B.
8738937	- Winch, U-90B.
10904960	- Winch, Planetary, P60.
10908544	- Winch and Controls, P30.
11641136	- Winch Assy Complete 25,000 lb Capacity, Right Input.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for winches. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.5.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.3 Construction. Winch construction shall be in accordance with the applicable model number, drawing number, and capacity listed in table I, as specified (see 4.5.2 and 6.2).

3.3.1 Adjustment mechanisms. All adjusting mechanisms shall function without interference, shall operate smoothly and shall maintain adjustment settings under all operating conditions of winch (see 4.5.2).

3.3.2 Pawl and ratchet. Unless otherwise specified (see 6.2), a ratchet and pawl arrangement shall be incorporated in the winch (see 4.5.2).

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3.3.3 Safety. All moving parts, so located that they may become a hazard to operating personnel, shall be fully enclosed or properly guarded. The torsional overload device furnished and installed shall prevent the exertion of a single line pull in excess of maximum single line pull indicated on applicable drawing. Winches conforming to Drawing 11641136 shall be in accordance with the single line pull as specified in table II. A minimum of a two to one safety factor shall be maintained for all components of the winch system (see 4.5.2).

3.3.4 Noise levels. If noise levels during operation of the winch exceed 85 dB(A) at the operator and occasionally occupied positions, then a hearing protection warning will be placed on the equipment where it is visible to the winch operator. The warning shall state that hearing protection is required during operation of the winch.

3.4 Performance.

3.4.1 Functional reliability. The winch operating under full load and at maximum input torque shall meet performance requirements for 20 hours without failure requiring replacement of parts. All winches shall operate throughout temperature range from -65 degrees Fahrenheit (°F) to +125°F without malfunction or damage to winch assembly (see 4.5.3.1, 4.5.3.2, and 6.5).

3.4.2 Winch speed and line pull. With winch operating at an input of 1800 revolutions per minute (rpm), the winches shall reel cable in and out and level wind at the maximum speed and line pull indicated on applicable drawing for winch specified in table I. Winch conforming to Drawing 11641136 shall meet the performance requirements specified in table II (see 4.5.3.1).

TABLE II. Functional requirements.

a.	Rated line pull-lbs (first layer on bare drum)	45 000 two part line
b.	Rated line pull-lbs (first layer on bare drum)	25 000 single line
c.	Line speed (based on 222 rpm input speed)	15 ft/min (1st layer)
d.	Torque - input shaft	8720 in. lbs
e.	Gear ratio	30 to 1
f.	Cable size	3/4 in. (herringbone constr.)
g.	Cable length	280 ft
h.	Cable travel shall be 15 rpm on first layer of drum.	
i.	When making a direct pull of 25 000 lb, pull requirements shall be obtained without parts replacement, drum deformation or winch failure.	

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TABLE II. Functional requirements - Continued.

j.	The winch shall withstand overload operation of 110 percent of the 25 000 rated single line pull for 10 minutes with no malfunction or damaged winch assembly.

3.4.3 Brake. The hydraulic, air operated, positive-action, friction-type brake shall control loads equal to maximum line pull as indicated on applicable drawing for winch specified in table I. The brakes shall stop and hold all loads up to and including the maximum specified on applicable drawing. When the winch controls are placed in neutral or hold position, the brakes shall be automatically actuated and applied (see 4.5.3.1).

3.4.4 Controls. The winch controls shall provide the winch operator with convenient and safe means of operating the winch (see 4.5.2 and 4.5.3).

3.5 Paint. The winch and parts which require painting shall be prepared and painted as specified in the contract or order (see 6.2). Color shall be as specified by the procuring activity (see 6.2). Painted surfaces shall be free from sags, runs, and thin areas (see 4.5.2).

3.6 Identification markings. Identification and markings of the fitting shall be permanent and legible and shall include, as a minimum (see 6.2 and 6.3):

a. Manufacturer's identification code (CAGE)
b. Date of manufacture
c. Part or identifying number (PIN).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one winch of each capacity when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table III), the materials examination (see 4.5.1) and the tests of 4.5.3.1 and 4.5.3.2 (see table IV).

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4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table III) and the tests of 4.5.3.1 (see table IV).

4.3.1 Sampling. Sampling for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

TABLE III. Classification of defects

Category	Defect	Method of examination
Critical	None	
Major:		
101	Winch line size capabilities (see 3.3).	Visual
102	Improper, poor assembly practice (see 3.3).	Visual
103	Dimension affecting interchangeability (see 3.3).	SIE 1/
104	Missing or inoperative components (see 3.3 and 3.3.3).	Visual and functional
105	Inferior quality (see 3.3).	Visual
106	Winch spooling malfunction (see 3.3.1 and 3.3.2).	Visual and functional
107	Adjustment and controls interference, rough operation, no response (see 3.3.1 and 3.4.4).	Functional
Minor:		
201	Incorrect type of lubrication (see 3.3).	Visual
202	Improper preparation for painting, color, application (see 3.5).	Visual
203	Incorrect or missing markings (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE IV. Classification of inspections

Title	Requirement	Inspec-tion	First article	Conformance	
				Examin-ation	Tests
Materials and construction	3.2	4.5.1	X		
Defects	3.3, 3.3.1, 3.3.2, 3.3.3, 3.5, and 3.6	4.5.2	X	X	
Functional reliability	3.4.1	4.5.3.1	X		X
Environmental	3.4.1	4.5.3.2	X		
Winch speed and line pull	3.4.2	4.5.3.1	X		X
Brake	3.4.3	4.5.3.1	X		X
Controls	3.4.4	4.5.3.1	X		X

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4.4 Examination. Each winch shall be examined for compliance with the requirements specified in 3.3 thru 3.3.3, 3.5 and 3.6. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3 thru 3.3.3, 3.5 and 3.6, shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, and by measurement with standard inspection equipment.

4.5.3 Performance.

4.5.3.1 Functional reliability test. To determine conformance to 3.4.1, 3.4.2, 3.4.3, and 3.4.4, the winch being tested shall be operated at speeds and with full load specified on applicable drawings for specific winch, using first layer on drum. Repeat complete cycles for 20 hours, 10 hours in low gear and 10 hours in high gear. During the test, a brake and overload test shall be accomplished by raising and lowering a load equal to 110 percent of full load specified on a specific drawing (see 3.3). The winch shall be operated for 5 minutes in low gear and 5 minutes in high gear, using the first layer of cable on the drum. The winch brakes shall stop and hold load twice during each upward and downward movement with a 5 foot (ft) interval between stops. All lines, levers and gears shall operate without binding or malfunction.

4.5.3.2 Environmental tests. To determine conformance to 3.4, the winch, serviced with applicable seasonal lubricant, shall be subjected to a temperature of -65°F for a period of 4 hours. The winch using first layer of cable on drum, shall be operated for 1 hour raising and lowering a full load for the specific winch specified in table I at a temperature of -60°F to -65°F. One-half of operation shall be in high gear, one-half hour in low gear. Load shall be stopped twice by winch brakes during each load lowering operation with a 5 ft interval between stops. The test procedure shall be repeated in temperatures from 120°F to 125°F with results recorded.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The winch, mounted on applicable military vehicle, is intended for use in pulling and hoisting heavy loads during recovery operations. Due to the extreme battlefield conditions under which this winch drum must operate (-60 to 125 °F), this item is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Model, class, capacity, PIN and applicable drawing of winch requirement (see 1.2 and 6.3).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.3).
d. If first article inspection is required (see 3.1).
e. Winch construction if other than as specified (see 3.3).
f. When pawl and ratchet are required (see 3.3.2).
g. Painting and color option (see 3.5).
h. Identification markings if other than as specified (see 3.6).
i. Packaging requirements (see 5.1).

6.3 Part identification number (PIN). The PINs to be used for semitrailers acquired to this specification are created as follows:

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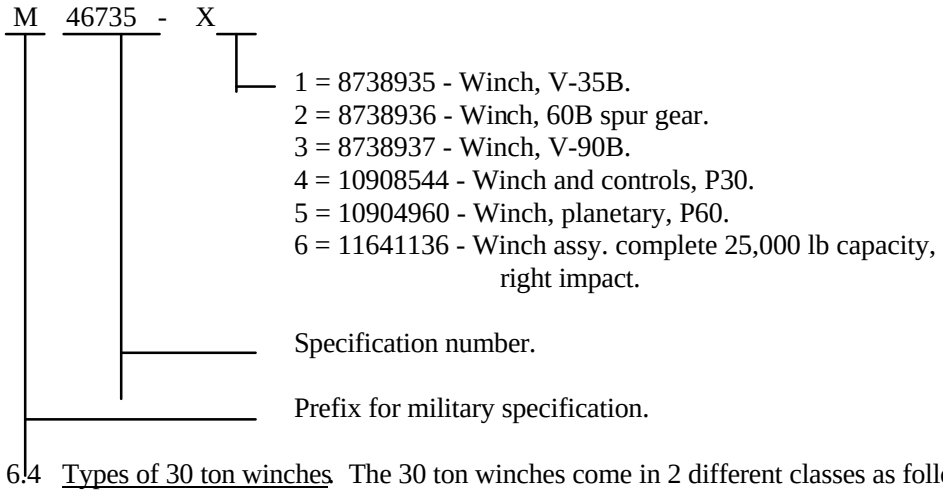

6.4 Types of 30 ton winches. The 30 ton winches come in 2 different classes as follows:

a. Drawing 8738936 - Winch, U-60B spur gear.
b. Drawing 10904960 - Winch, P60 planetary gear.

6.5 Recommended service products. Typically, the winch should be serviced with MIL-PRF-2105 or equivalent 80W-90 oil for temperatures ranging from 125 to -10°F and grade 75W for temperatures from -10 to -65°F.

6.6 Subject term (key word) listing

Geared
Hydraulic system
Planetary
Power operated spur
Tactical
Transport

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
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Preparing Activity:
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL46735C(AT)	**2. DOCUMENT DATE (YYMMDD)** 980831

3. DOCUMENT TITLE WINCH DRUM: VEHICLE MOUNTING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

NAME COMMANDER US ARMY TANK-AUTOMOTIVE AND ARMAMENTS COMMAND	b. TELEPHONE *Include Area Code)* (1) Commercial 810-574-8745 (2) AUTOVON 786-8745
c. ADDRESS *(Include Zip Code)* ATTN: AMSTA-TR-E/BLUE WARREN, MI 48397-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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